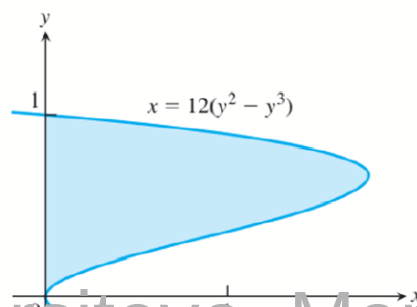


Q 1. Use cylindrical shells to find the volume of the solid that is generated when the region enclosed by $y = x^3$, $y = 1$, $x = 0$ is revolved about the line $y = 1$.

Q 2. Use shell method to find the volumes of the solids generated by revolving the shaded regions about the indicated axes

a) the x -axis

b) the line $y = 1$.



Q 3. Find the length of the following curves described by the functions

a) $24xy = y^4 + 48$ from $y = 2$ to $y = 4$.

b) $y = \int_0^x \sqrt{1-t^2} dt$, $0 \leq x \leq 1$.

Q 4. Find the arc length function for the graph of $f(x) = 2x^{3/2}$ using $(0,0)$ as the starting point. What is the length of the curve from $(0,0)$ to $(1,2)$?

Q 5. Find the area of the surfaces obtained by rotating the curve

a) $x = 1 + 2y^2$, $1 \leq y \leq 2$ about the x -axis.

b) $y = (x^2/2) + (1/2)$, $0 \leq x \leq 1$ about the y -axis.

Q 6. Let C denote the curve $y = 2 - |x|$, $-1 \leq x \leq 1$.

a) Compute the volume of the solid of revolution formed by revolving the region bounded between C and the x -axis about both x -axis and y -axis separately.

b) Compute the surface area of the surface of revolution of C about the x -axis.

c) Compute the arc length of C .